

Data Integrity Report

1. Missing Values

What was found

The combined dataset contained missing values in 17 columns. The missing values did not all have the same cause, so each column was considered separately rather than applying one treatment to all missing values.

Where it was

Missing values were found in these columns:

first_name, last_name, grade, neighborhood, membership_status, join_date, checkout_id, book_id, checkout_date, return_date, book_count, title, author, genre, pages, publication_year, publisher

How big it was

The number of missing values in each column was:

first_name — 5 missing values: These records belong to unregistered member IDs, so their names were not available in the members table. I left these values missing because the names cannot be reliably determined from the available data.

last_name — 5 missing values: These records belong to unregistered member IDs, so their last names were not available. I left these values missing because they cannot be reliably determined.

grade — 43 missing values: Some registered members had no grade recorded, while some Reading Kickoff records belonged to unregistered members. I filled the missing values using the mode, because grade is a categorical/discrete value and the most common grade provides a reasonable estimate while keeping the values within the grades already present in the dataset.

neighborhood — 5 missing values: Only 5 values were missing. I filled them using the mode, because neighborhood is a categorical variable and the small number of missing values makes mode imputation a reasonable choice.

membership_status — 5 missing values: These records belong to unregistered member IDs, so their membership status could not be determined. I left these values missing rather than assigning a status without reliable evidence.

join_date — 12 missing values: I left these values missing because there was no reliable date that could be used to fill them.

checkout_id — 44 missing values: 18 belonged to registered members who had no checkouts, while the other 26 came from Reading Kickoff records, which did not contain a checkout ID column. I left these values missing because a checkout ID cannot be estimated.

book_id — 18 missing values: These records belonged to registered members who had no checkout records. Since there was no associated book, I left the values missing.

checkout_date — 18 missing values: These records belonged to members with no checkout records, so there was no checkout date to provide. I left the values missing.

return_date — 109 missing values: Some records had checkouts but no recorded return date, while others had no checkout or came from Reading Kickoff records that did not contain return dates. I left these values missing because the actual return dates could not be reliably determined.

book_count — 26 missing values: These values belonged to Reading Kickoff records. Since these members were not included in the database checkout records, their total number of borrowed books could not be determined directly. I left the values missing rather than assigning an estimated count.

title — 18 missing values: These records had no associated book information because they had no book ID. I left the values missing because there was no correct title that could be determined.

author — 18 missing values: These records had no associated book information. I left the values missing because the correct author could not be determined.

genre — 18 missing values: These records had no associated book information. I left the values missing because the correct genre could not be determined.

pages — 18 missing values: These records had no associated book information. I left the values missing rather than assigning an estimated page count to a record with no identified book.

publication_year — 53 missing values: Some records had no associated book, while others had book information but no recorded publication year. I left the values missing rather than assigning an estimated publication year to a record that may be without an identified book.

publisher — 18 missing values: These records had no associated book information, so their publishers could not be determined. I left these values missing rather than assigning the most common publisher to unrelated records.

What was done and why

The missing values were handled column by column according to what each column represented. Two columns were suitable for statistical imputation: neighborhood and grade were filled using their modes.

Fields such as names, IDs, dates, book details, and membership status were not statistically filled because doing so could introduce information that was not supported by the original data.

2. Duplicate Records

What was found

The dataset contained **8 true duplicate records**. Although checking duplicates using only `checkout_id` is the logical approach, but this column already has a lot of nans (because this column wasn't in the reading kickoff file), and these nans will be calculated as duplicates. So `checkout_id` alone was not used to determine whether records were genuinely duplicated.

The dataset was instead checked across **all columns** to identify records that were completely identical, and to ensure it's accurate, I subtracted the length of the reading kickoff rows (no. of nans in `checkout_id`) from the no. of duplicates in `checkout_id`, and the output was as same as no. of duplicates across all columns, confirming all the extra duplicates in `checkout_id` are nans.

Where it was

The issue occurred across the combined dataset rather than in a specific column.

How big it was

There were **8 complete duplicate records**.

What was done and why

The 8 true duplicate records were removed using:

`df = df.drop_duplicates()`. This was appropriate because the records were identical across all columns and therefore represented repeated copies of the same record.

Records that were merely similar but contained different information were not removed.

3. Inconsistent Text Values

What was found

Several text columns contained values that could potentially be written in different forms, such as differences in capitalization and extra spaces. For example, the dataset contained membership statuses such as Active, active, Inactive, and INACTIVE, as well as neighborhood names with inconsistent capitalization.

Where it was

The text columns with inconsistency problems:

- neighborhood
- membership_status

However, I applied the strip and title functions to all columns, because there was no way to know which columns are inconsistent except by reading them manually, which can lead to mistakes.

How big it was

Neighborhood: 4 redundant unique forms

Membership_status: 3 redundant unique forms

What was done and why

The text values were standardized using:

```
df[column] = df[column].str.strip().str.title()
```

This removes unnecessary spaces at the beginning or end of values and changes capitalization to a consistent title-case format.

For example, different capitalization forms of the same membership status are brought to a single form:

Active, active, and ACTIVE → Active

Similarly, neighborhood names with inconsistent capitalization are standardized.

This was done to make different written forms of the **same real-world value** consistent without intentionally combining genuinely different values.

Note: There were no other inconsistencies that needed treating.

4. Unregistered Member IDs

What was found

Some records in the combined dataset contained `member_id` values that did not exist in the registered members table.

Where it was

The issue occurred in the `member_id` column.

The invalid records were identified by comparing the `member_id` values in the combined dataset with the registered IDs in `df_members`:

```
unregistered_members = df[
    ~df['member_id'].isin(df_members['member_id'])
]
```

How big it was

There were **5 records** referencing member IDs that were not present in the registered members table.

What was done and why

All records containing these unregistered member IDs were removed:

```
df = df[df['member_id'].isin(df_members['member_id'])]
```

This decision was applied consistently to all unregistered-member records.

The registered members table itself was not modified. Removing these records prevents information from being incorrectly associated with registered members when there is no matching member record to support that association.
